



CELEBAL
TECHNOLOGIES

**DELTA LAKE INCREMENTAL DATA PROCESSING
USING PYSPARK**

Celebal Summer Internship 2026

Week 7 Assignment

**Delta Lake Incremental Data Processing Using
MERGE Operation**

Dataset: Sample Superstore Dataset

Submission Date:

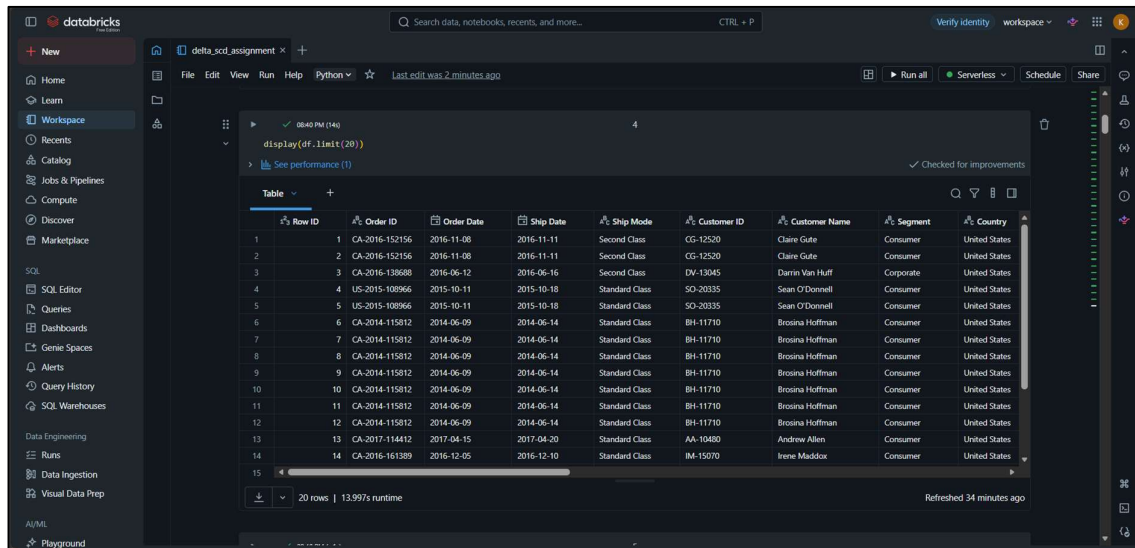
05-07-2026

Prepared By:

Keshav Kumar sharma

Loading and Displaying the Customer Dataset

The Sample Superstore dataset was successfully loaded into a PySpark DataFrame and the first 20 records were displayed to verify that the data was imported correctly.

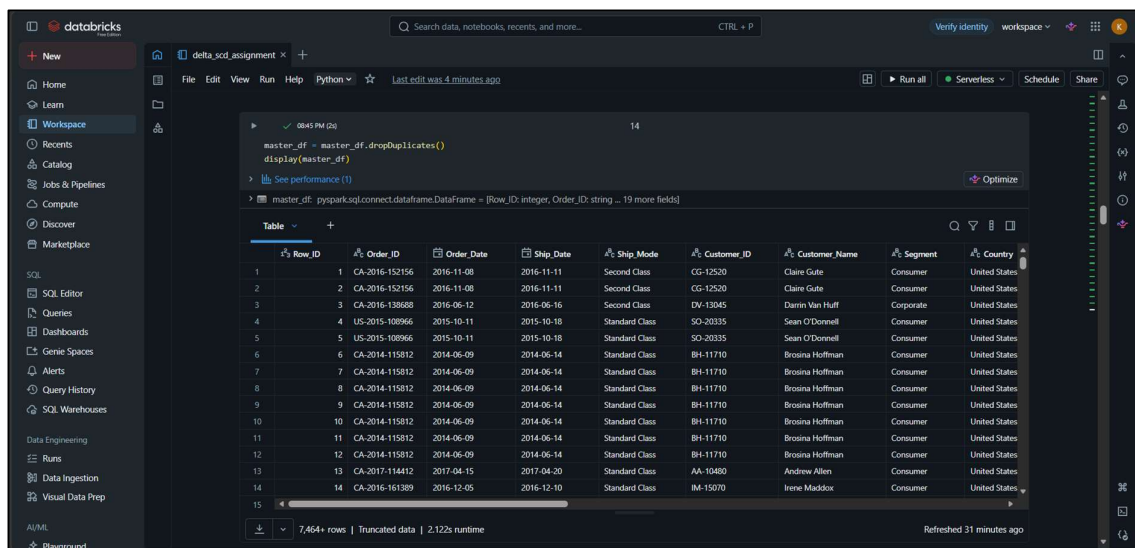


The screenshot shows the Databricks workspace interface. The notebook code cell contains the command `display(df.limit(20))`. The output is a table with 20 rows and 9 columns: Row ID, Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Customer Name, Segment, and Country. The data is as follows:

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country
1	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gule	Consumer	United States
2	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gule	Consumer	United States
3	CA-2016-138688	2016-06-12	2016-06-16	Second Class	DV-13045	Darrin Van Huff	Corporate	United States
4	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States
5	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States
6	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
7	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
8	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
9	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
10	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
11	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
12	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
13	CA-2017-114412	2017-04-15	2017-04-20	Standard Class	AA-10480	Andrew Allen	Consumer	United States
14	CA-2016-161389	2016-12-05	2016-12-10	Standard Class	IM-15070	Irene Maddox	Consumer	United States

Step 2: Data Cleaning and Removing Duplicate Records

Duplicate records were removed from the dataset using the `dropDuplicates()` function to ensure data consistency before storing the data in the Delta Table.



The screenshot shows the Databricks workspace interface. The notebook code cell contains the commands `master_df = master_df.dropDuplicates()` and `display(master_df)`. The output is a table with 14 rows and 9 columns: Row ID, Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Customer Name, Segment, and Country. The data is as follows:

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country
1	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gule	Consumer	United States
2	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gule	Consumer	United States
3	CA-2016-138688	2016-06-12	2016-06-16	Second Class	DV-13045	Darrin Van Huff	Corporate	United States
4	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States
5	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States
6	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
7	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
8	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
9	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
10	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
11	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
12	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
13	CA-2017-114412	2017-04-15	2017-04-20	Standard Class	AA-10480	Andrew Allen	Consumer	United States
14	CA-2016-161389	2016-12-05	2016-12-10	Standard Class	IM-15070	Irene Maddox	Consumer	United States

Step 3: Data Validation and Data Type Conversion

The cleaned dataset was validated by checking the total number of records after removing duplicates. The Sales column was converted to the Double data type to ensure compatibility with Delta Lake operations and accurate numerical processing.

The screenshot shows the Databricks workspace interface. The left sidebar contains navigation options like Home, Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Discover, Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie Spaces, Alerts, Query History, SQL Warehouses, Data Engineering, Runs, Data Ingestion, Visual Data Prep, AI/ML, and Playground. The main area displays a notebook with the following code and output:

```
master_df = master_df.withColumn("Sales", col("Sales").cast("double"))
display(master_df.limit(10))
```

The output shows a table with 10 rows and 10 columns: Row_ID, Order_ID, Order_Date, Ship_Date, Ship_Mode, Customer_ID, Customer_Name, Segment, and Country. The first 10 rows are displayed.

Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Customer_Name	Segment	Country
1	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12620	Claire Gule	Consumer	United States
2	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gule	Consumer	United States
3	CA-2016-130688	2016-06-12	2016-06-16	Second Class	DV-13045	Darrin Van Huff	Corporate	United States
4	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States
5	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States
6	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
7	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
8	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
9	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
10	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States

Step 4: Updating Existing Customer Records

Existing customer records were updated by modifying the Order_ID, Customer_Name, and Sales values for selected Row_IDs. The updated records were displayed to verify that the changes were applied successfully before creating the incremental dataset.

The screenshot shows the Databricks workspace interface. The left sidebar contains navigation options like Home, Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Discover, Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie Spaces, Alerts, Query History, SQL Warehouses, Data Engineering, Runs, Data Ingestion, Visual Data Prep, AI/ML, and Playground. The main area displays a notebook with the following code and output:

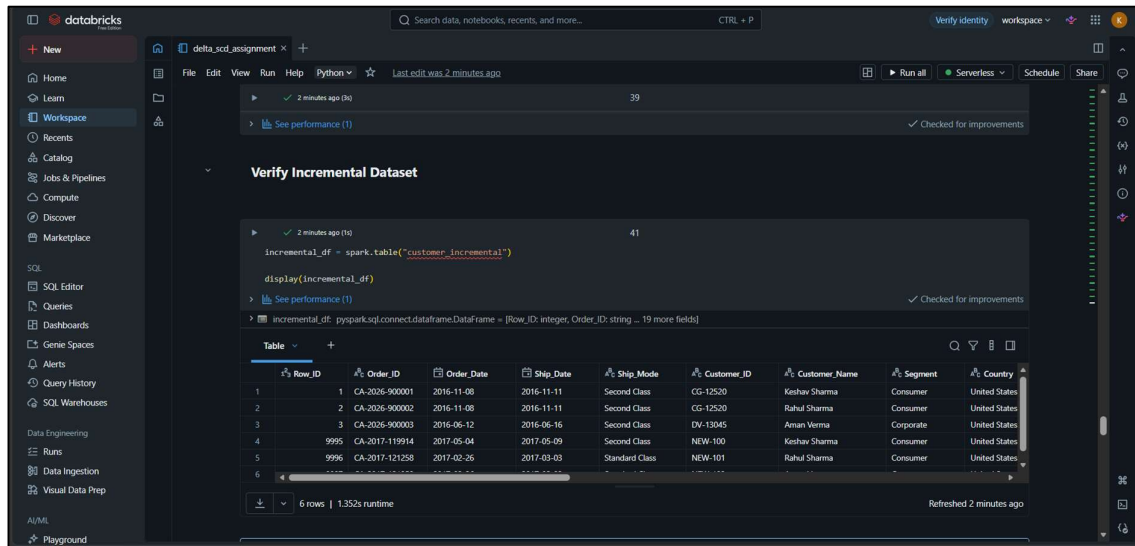
```
updated_df = pyspark.sql.connect.dataframe.DataFrame([Row_ID: integer, Order_ID: string ... 19 more fields])
display(updated_df.select("Row_ID", "Sales", "Customer_Name"))
```

The output shows a table with 3 rows and 3 columns: Row_ID, Sales, and Customer_Name. The first 3 rows are displayed.

Row_ID	Sales	Customer_Name
1	361.96	Keshav Sharma
2	831.94	Rahul Sharma
3	114.62	Aman Verma

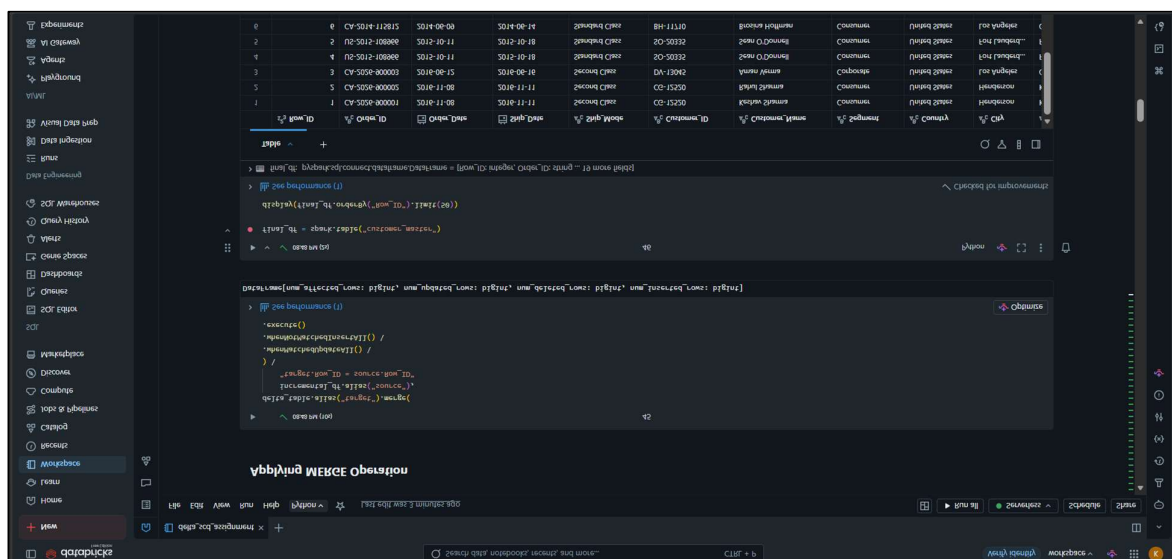
Step 5: Creating and Verifying the Incremental Dataset

An incremental dataset containing both updated records and new customer records was created and stored as a Delta Table. The dataset was then displayed to verify that all changes were correctly prepared before performing the MERGE operation.



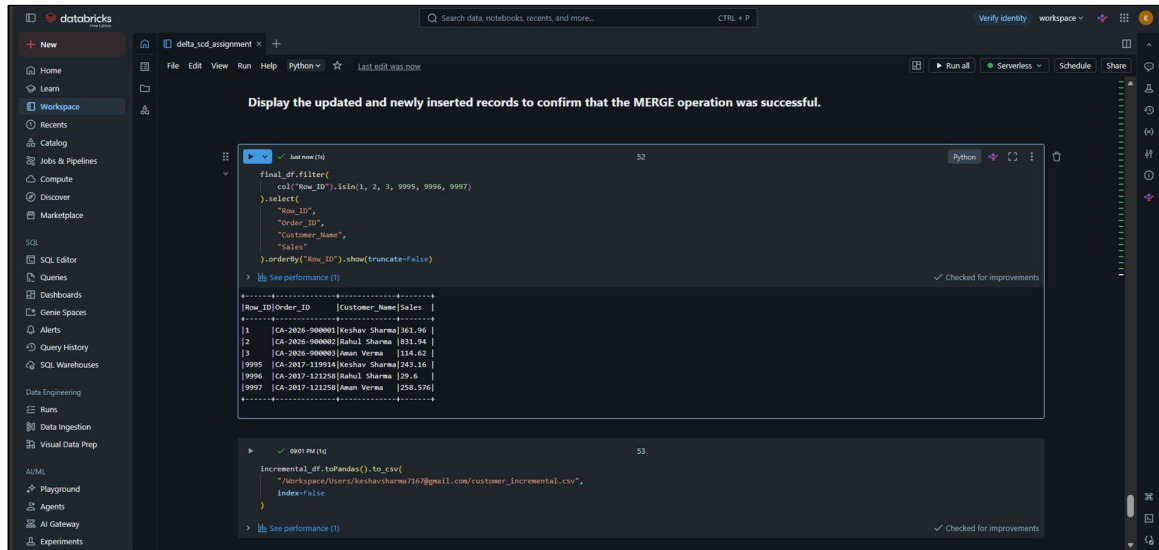
Step 6: Applying the Delta Lake MERGE Operation

The **MERGE** operation was performed on the Delta Table using the incremental dataset. Existing records were updated, new records were inserted, and the final Delta Table was displayed to verify that the merge was executed successfully.



Step 7: Validating the Final MERGE Results

The final Delta Table was validated by displaying both the updated records (Row_ID 1, 2, 3) and the newly inserted records (Row_ID 9995, 9996, 9997). The output confirms that the MERGE operation successfully updated existing records and inserted new records without creating duplicate Row_IDs.



The screenshot shows the Databricks workspace interface. The notebook is titled "delta_scd_assignment" and contains a Python cell with the following code:

```
final_df.filter(
    col("Row_ID").isin(1, 2, 3, 9995, 9996, 9997)
).select(
    "Row_ID",
    "Order_ID",
    "Customer_Name",
    "Sales"
).orderBy("Row_ID").show(truncate=False)
```

The output of the code is a table with 6 rows, showing the updated and newly inserted records:

Row_ID	Order_ID	Customer_Name	Sales
1	CA-2026-900001	Keshav Sharma	361.96
2	CA-2026-900002	Rahul Sharma	831.94
3	CA-2026-900003	Aman Verma	114.62
9995	CA-2027-119914	Keshav Sharma	243.16
9996	CA-2027-121258	Rahul Sharma	129.6
9997	CA-2027-121258	Aman Verma	258.576

The notebook also shows a second cell with the following code:

```
Incremental_df.toPandas().to_csv(
    "/Workspace/Users/keshavsharma716@gmail.com/customer_incremental.csv",
    index=False
)
```